Primary Six Mathematics — Finding unknown prime factors

Teacher answer key

Worked examples

1. Two numbers are 12 and x. Their HCF is 6 and LCM 60. Find x.

- 6×60=12×x; x=30. Check: HCF(12,30)=6 and LCM(12,30)=60.

2. If 36=2^2×3^2, which prime factor is missing in 2×2×3×__?

- 3.

Practice answers

1. Find x if HCF=3, LCM=30, a=6. — x = (3 × 30) ÷ 6 = 15. Check: HCF(6,15)=3 and LCM(6,15)=30. Both conditions hold.

2. Find x if HCF=5, LCM=60, a=15. — x = (5 × 60) ÷ 15 = 20. Check: HCF(15,20)=5; LCM(15,20)=60.

3. Missing factor: 30=2×3×__. — 5

4. Missing factor: 84=2×2×3×__. — 7

5. HCF of 18 and 27. — 9

6. LCM of 18 and 27. — 54

7. HCF×LCM for 8 and 12. — 96

8. Find b if HCF=4, LCM=40, a=8. — b = (4 × 40) ÷ 8 = 20. Check: HCF(8,20)=4; LCM(8,20)=40.

Marking: Accept correct equivalent methods and simplified answers.